

CHAPTER 5: TWO-DIMENSIONAL ANALYTICAL GEOMETRY-II

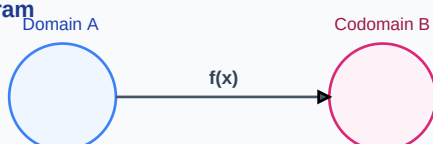
PREMIUM ACADEMIC STUDY SUITE & EXAM GUIDE

Board:	Tamil Nadu State Board (TNSB)
Syllabus:	Samacheer Kalvi Textbook Curriculum aligned
Standard:	Class 12 English Medium
Subject:	Mathematics Volume 1
Chapter Name:	Two-Dimensional Analytical Geometry-II

Learning Objectives:

- Comprehend core theoretical concepts and exact textbook terminology.
- Apply relevant formulas and proofs to solve high-scoring board problems.
- Interpret and sketch vector diagrams or graphical models representing equations.
- Build examination readiness for 1-mark, 2-mark, and 5-mark question sets.

Function Mapping Diagram



Core Concepts & Definitions

Textbook definitions are essential for scoring fully in the short-answer section of board assessments. Review the primary definitions below:

Term	Official Definition & Context
Parabola	The locus of a point which moves such that its distance from a fixed point (focus) equals its distance from a line (directrix).
Ellipse	The locus of a point whose sum of distances from two fixed points (foci) is a constant equal to major axis length.
Asymptotes of Hyperbola	A pair of straight lines passing through the center of a hyperbola that the curve approaches infinitely close but never touches.

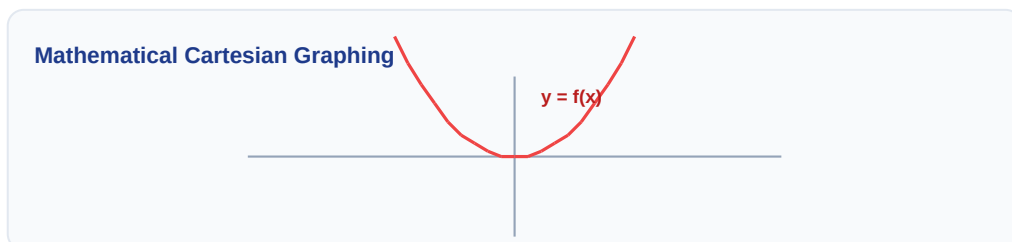
Syllabus Exam Tip: Memorize precise keywords. Examiners award maximum marks when textbook terms and correct symbols are correctly referenced.

Venn Diagram: $A \cap B$



Detailed Explanation & Formulas

Here, we analyze the core laws and formulas. Examine the conceptual illustration below to visual the relation between variables:



Essential Formulas, Laws & Identities:

- Parabola ($y^2 = 4ax$): Focus $(a, 0)$ | Directrix $x = -a$
- Ellipse ($x^2/a^2 + y^2/b^2 = 1$): Foci $(\pm ae, 0)$ | Eccentricity $e = \sqrt{1 - b^2/a^2}$
- Hyperbola ($x^2/a^2 - y^2/b^2 = 1$): Foci $(\pm ae, 0)$ | Eccentricity $e = \sqrt{1 + b^2/a^2}$
- Tangent condition: For $y = mx + c$ to touch $y^2 = 4ax$, $c = a/m$

Make sure to check the dimensional consistency of all physical quantities and verify that units are correctly stated in final solutions.

Worked Examples

Review these standard board examination-style worked examples to learn proper step layout structures:

Example 1: Find the coordinates of focus and length of latus rectum of parabola $y^2 = 12x$.

Step-by-step Solution:

Comparing with $y^2 = 4ax$, $4a = 12 \Rightarrow a = 3$.

Focus: $(a, 0) = (3, 0)$.

Length of latus rectum: $4a = 12$.

Linear Algebraic Matrix Representation

$$\begin{bmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \end{bmatrix}$$

HOTS (Higher Order Thinking Skills) Challenge

Challenge yourself with these complex, multi-concept problems designed to test deeper conceptual understanding and mathematical reasoning:

HOTS Challenge 1: Find equation of tangent and normal to the ellipse $x^2/16 + y^2/9 = 1$ at point $(2, 3\sqrt{3}/2)$.

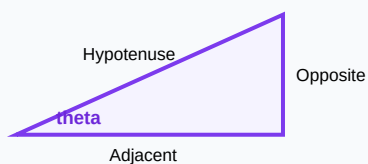
Analytical Solution Strategy:

Equation of tangent: $x \cdot x_1/a^2 + y \cdot y_1/b^2 = 1$.

Substitute (x_1, y_1) : $2x/16 + [y \cdot 3\sqrt{3}/2]/9 = 1 \Rightarrow x/8 + \sqrt{3}y/6 = 1 \Rightarrow 3x + 4\sqrt{3}y - 24 = 0$.

Normal slope is perpendicular to tangent: normal eq is $4\sqrt{3}x - 3y - 5\sqrt{3} = 0$.

Right-Angled Triangle Geometry



Practice Worksheet

Test your skills by attempting these sample exam problems independently. Draw diagrams where appropriate:

Q1 (1-Mark): Find eccentricity of the ellipse $9x^2 + 25y^2 = 225$.

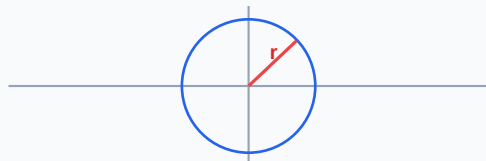
Q2 (2-Mark): Write standard equation of hyperbola with foci $(\pm 5, 0)$ and transverse axis length 8.

Q3 (2-Mark): State the equation of asymptotes of hyperbola $x^2/a^2 - y^2/b^2 = 1$.

Q4 (5-Mark): Find length of latus rectum of ellipse $x^2/9 + y^2/4 = 1$.

Q5 (5-Mark): Determine the condition for line $y = mx + c$ to touch ellipse $x^2/a^2 + y^2/b^2 = 1$.

Circle with Radius Vector on Plane



Real-World Applications & Connections

This chapter connects directly with modern industrial engineering, computation, and natural systems in numerous ways:

Application Case: Acoustic Whispering Galleries

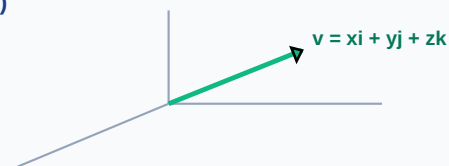
Elliptical ceilings reflect sound waves from one focal point directly to the opposite focal point.

Application Case: Satellite Dish design

Parabolic reflectors converge parallel incoming microwave signals into a central focus receiver.

Cross-Curricular Link: Concepts learned in this unit are heavily applied in other subjects like Physics (equations of motion, field vectors) and Data Analytics.

3D Vector Projections (x, y, z)



Revision Summary & Strategy

Use this checklist to self-evaluate your exam preparation status before final tests:

Key Recall Tips:

- ✓ Always write down 'Given Parameters' first to ensure partial marks are locked in.
- ✓ Pay attention to signs (+ / -) when performing algebraic operations or force mappings.
- ✓ Check that you have written the correct units (e.g. Joules, Tesla, cm^3 , N) for the final numeric answer.
- ✓ Draw neat diagrams using a sharp pencil and ruler for geometry or circuit layouts.

Syllabus Checklist:

- ☐ Read and memorized core definitions on Page 2.
- ☐ Reviewed critical formulas and relations on Page 3.
- ☐ Solved worked examples on Page 4.
- ☐ Attempted worksheet problems on Page 6.

CONGRATULATIONS! YOU HAVE COMPLETED THIS SYLLABUS STUDY GUIDE!

Practice consistently, verify answers with standard textbook methods, and take the topic test in the Nexora Learning app to gauge progress.
Good luck!

Probability Tree Outcome Diagram

